



RedHawk Vision's Paparazzi software generates great stills from video

The Moon Again

The abovementioned technologies, which NASA helped develop for use in space exploration, have already been put to practical use in products that we can go out and buy. We haven't even scratched the surface yet, and providing any sort of detailed analysis would fill 98,16,345 issues of Digit! However, let's take a look at NASA's plans for a return to the Moon.

An article titled "How We'll Get Back To The Moon" on NASA's site says:

"A heavy-lift rocket blasts off, carrying a lunar lander and a "departure stage" needed to leave Earth's orbit. The crew launches separately, then docks their capsule with the lander and departure stage and heads for the Moon. Three days later, the crew goes into lunar orbit. The four astronauts climb into the lander, leaving the capsule to wait for them in orbit. After landing and exploring the surface for seven days, the crew blasts off in a portion of the lander, docks with the capsule and travels back to Earth. After a de-orbit burn, the service module is jettisoned, exposing the heat shield for the first time in the mission. The parachutes deploy, the heat shield is dropped and the capsule sets down on dry land."

Those were the details of how the first mission to the Moon will be carried out. The article continues, and explains what will happen next:

"With a minimum of two lunar missions per year, momentum will build quickly toward a permanent outpost. Crews will stay longer and learn to exploit the Moon's resources, while landers make one way trips to deliver cargo. Eventually, the new system could rotate crews to and from a lunar outpost every six months. Planners are already looking at the lunar south pole as a candidate for an

outpost because of concentrations of hydrogen thought to be in the form of water, ice, and an abundance of sunlight to provide power.

"These plans give NASA a huge head start in getting to Mars. We will already have the heavy-lift system needed to get there, as well as a versatile crew capsule and propulsion systems that can make use of Martian resources. A lunar outpost just three days away from Earth will give us needed practice of "living off the land" away from our home planet, before making the longer trek to Mars."

Mars or not, we've already seen how NASA's research is improving the quality of life right here on Earth—and we're not complaining. You should pay www.nasa.gov a visit and take a look at the thousands of pictures, videos and articles on current research and their plans for conquering space.

Coming Down To Earth

A lot of what's been said here may seem very futuristic; to some, perhaps, it may border on fiction. However, scientists and explorers for generations have surprised us and made the cynics eat humble pie for lunch and dinner!

The facts are that research is a never-ending phenomenon at places like NASA, and the success of projects that may have seemed incredible just a decade ago are already being used in the products we mentioned!

When the space explorers talk, we better listen, because space agencies are sending out one clear signal: for the betterment of mankind, space exploration and research is truly The Final Frontier! ■

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Images courtesy NASA and Spinoff Magazine

